$$P_{01} = \frac{\sum\left[\frac{p_1}{p_0} \times 100\right] \times p_0 q_0}{\sum p_0 q_0} = P_{01} = \frac{\sum wp}{\sum w}$$

The weighted average price relatives using geometric mean:

$$P_{01} = \text{antilog}\left(\frac{\sum w \log p}{\sum w}\right)$$

Example 6.11

Compute price index for the following data by applying weighted average of price relatives method using (i) Arithmetic mean and (ii) Geometric mean.

Item	p_0	q_0	p_1
Wheat	3.0	20 kg	4.0
Flour	1.5	40 kg	1.6
Milk	1.0	10 kg	1.5

Solution:

(i) Computation for the weighted average of price relatives using arithmatic mean.

Item	p_0	q_0	p_1	w	p	$\log p$	wp	$w \log p$
Wheat	3.0	20	4.0	60	133.3	2.1249	7998	127.494
Flour	1.5	40	1.6	60	106.7	2.0282	6402	121.692
Milk	1.0	10	1.5	10	150.0	2.1761	1500	21.761
				$\sum w = 130$			$\sum wp = 15900$	$\sum w \log p = 270.947$

$$P_{01} = \frac{\sum wp}{\sum w} = \frac{15,900}{130} = 122.31$$

This means that there has been a 22.31 % increase in prices over the base year.

(ii) Index number using geometric mean of price relatives is:

$$P_{01} = \text{Antilog}\ \frac{\sum w \log p}{\sum w} = \text{Antilog}\ \frac{270.947}{130}$$

$$= \text{Antilog}\ (2.084) = 121.3$$

This means that there has been a 21.3 % increase in prices over the base year.

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